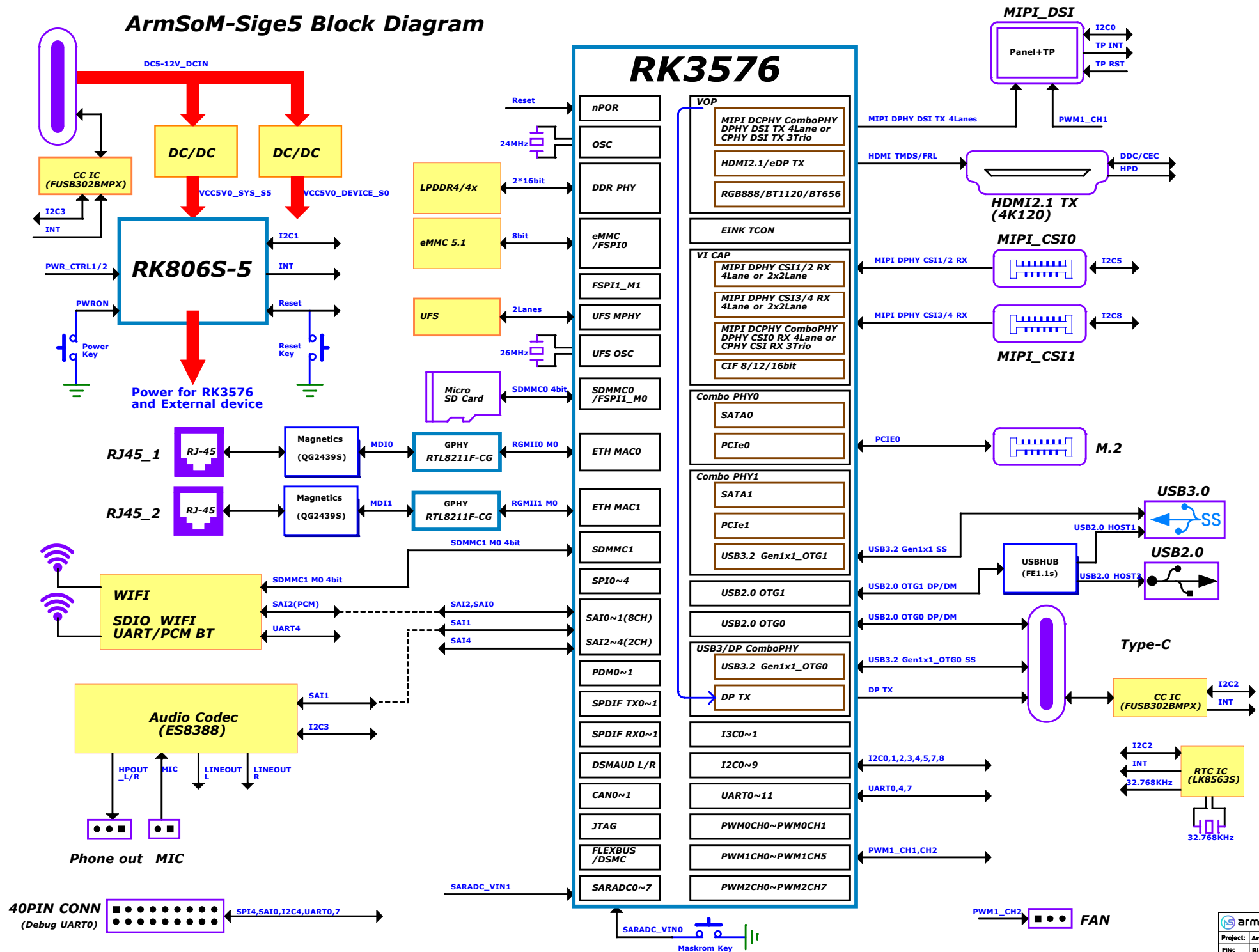
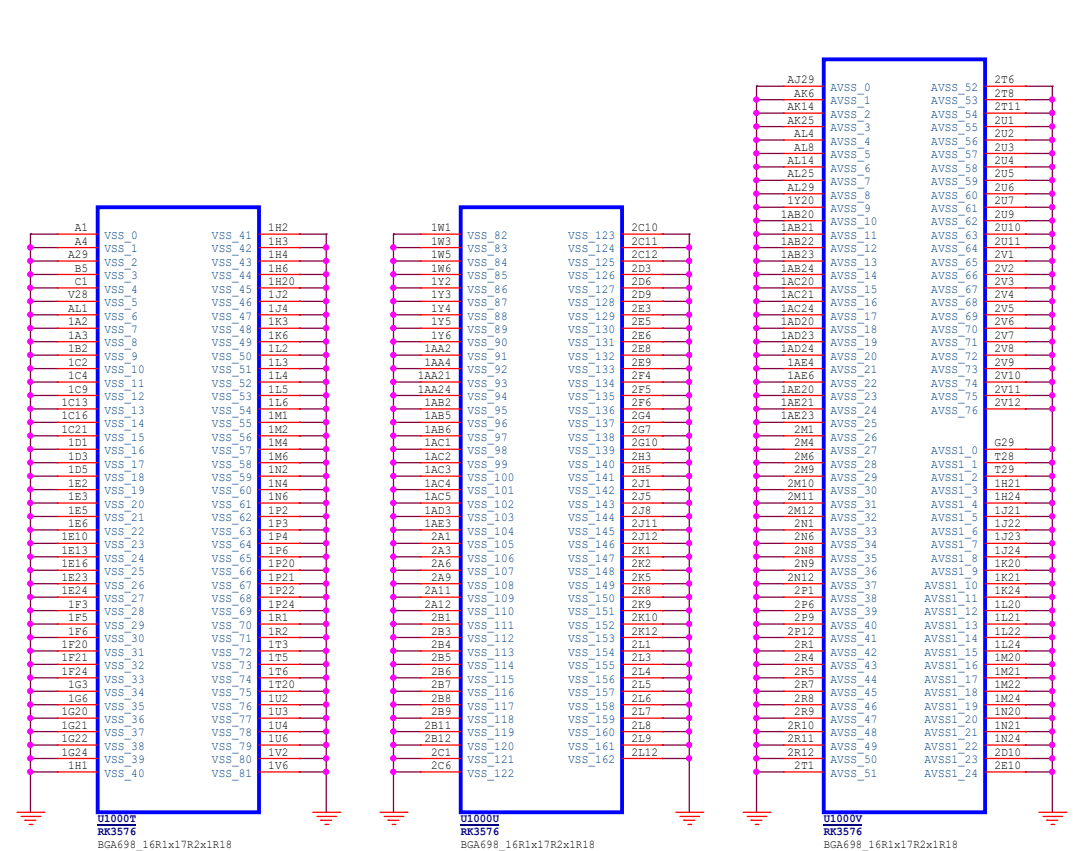
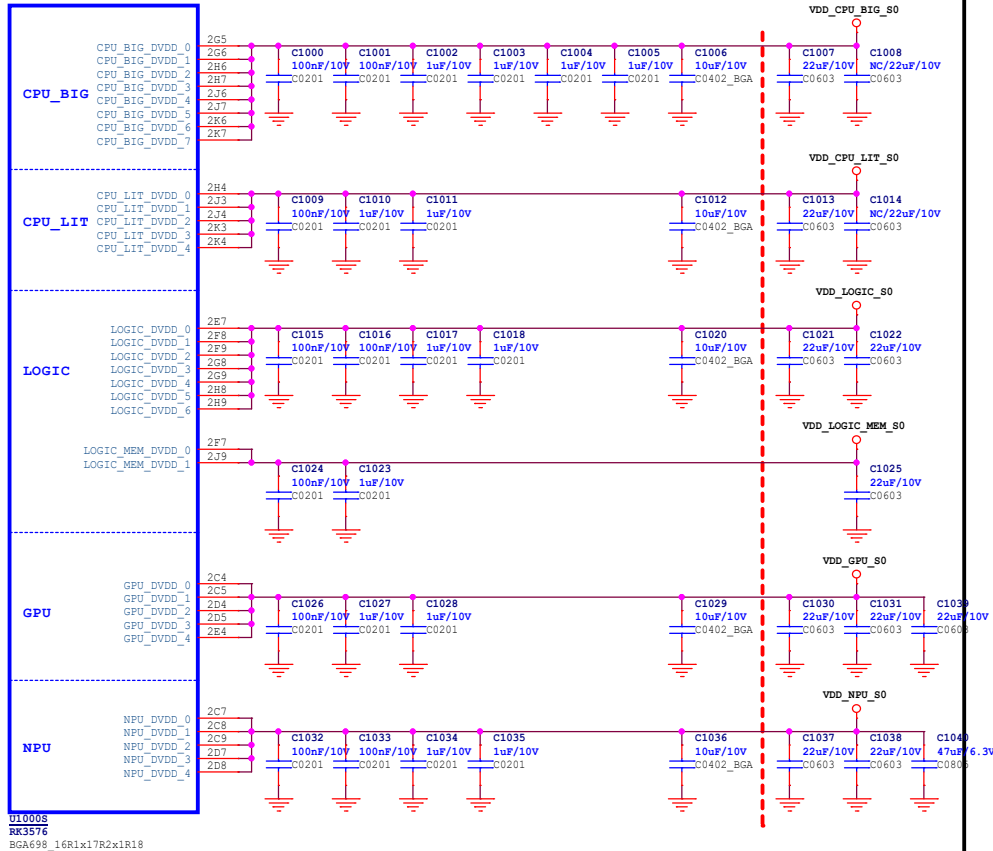


ArmSoM-Sige5 Block Diagram

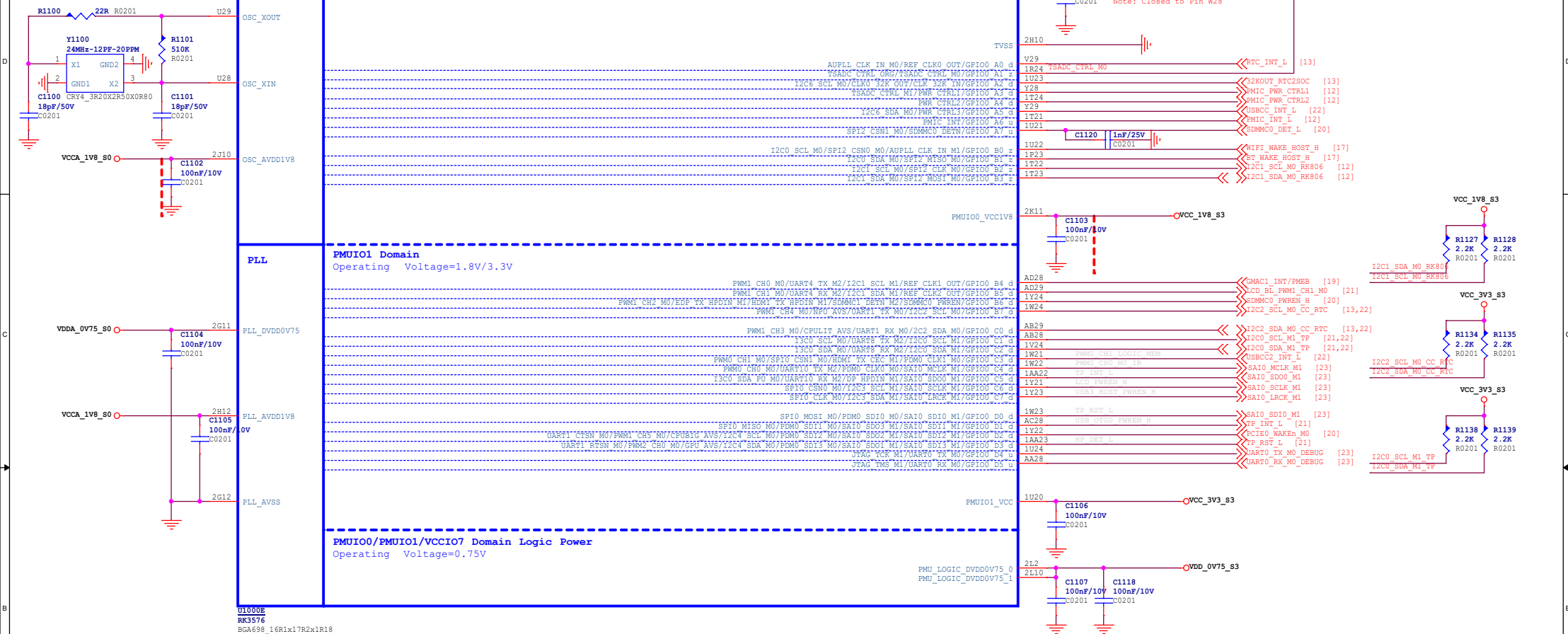


RK3576_S (Power)

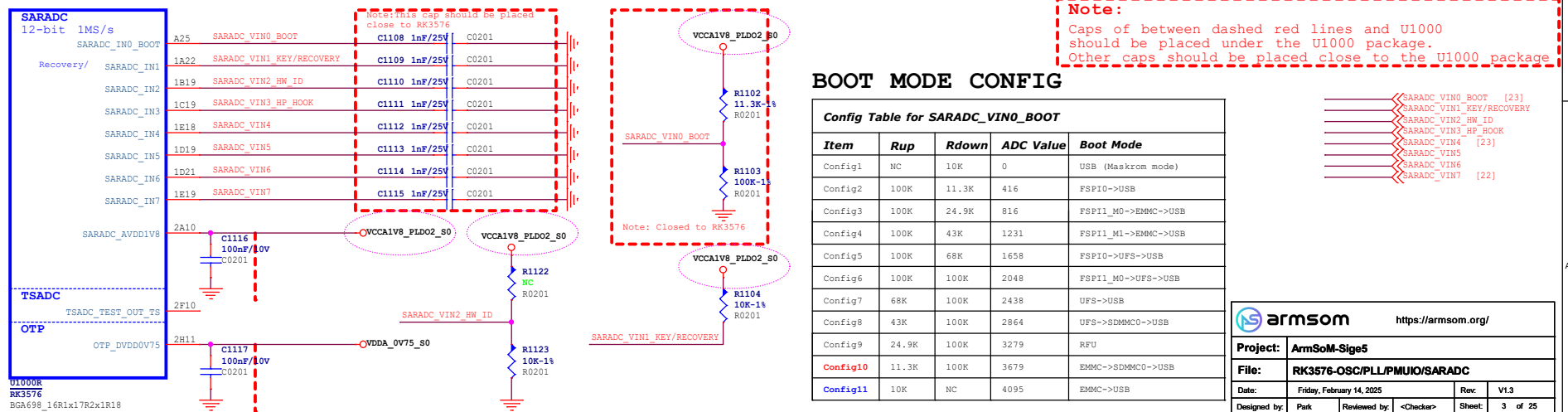
RK3576_T/U/V (GND)



RK3576_E
(PMUIO $\overline{0}$ /1)



RK3576_R
(SARAD $\bar{C}$)



RK3576_A (DDRPHY)

LPDDR4 Signal

DDR FILTER

Note1: Caps in the red line dotted box should be placed under the U1000 package

Note2: Resistors in the red line dotted box should be placed under the U1000 package

Note: (1) Power Sequence: VDD-VDDQ_CKE-VDDQ (2) Hold power of DDRPHY_CKE_VDDQ during retention times.

Note1: Caps in the red line dotted box should be placed under the U1000 package

Note2: Resistors in the red line dotted box should be placed under the U1000 package

Project: ArmSoM-Sigs5

File: RK3576-DDR PHY

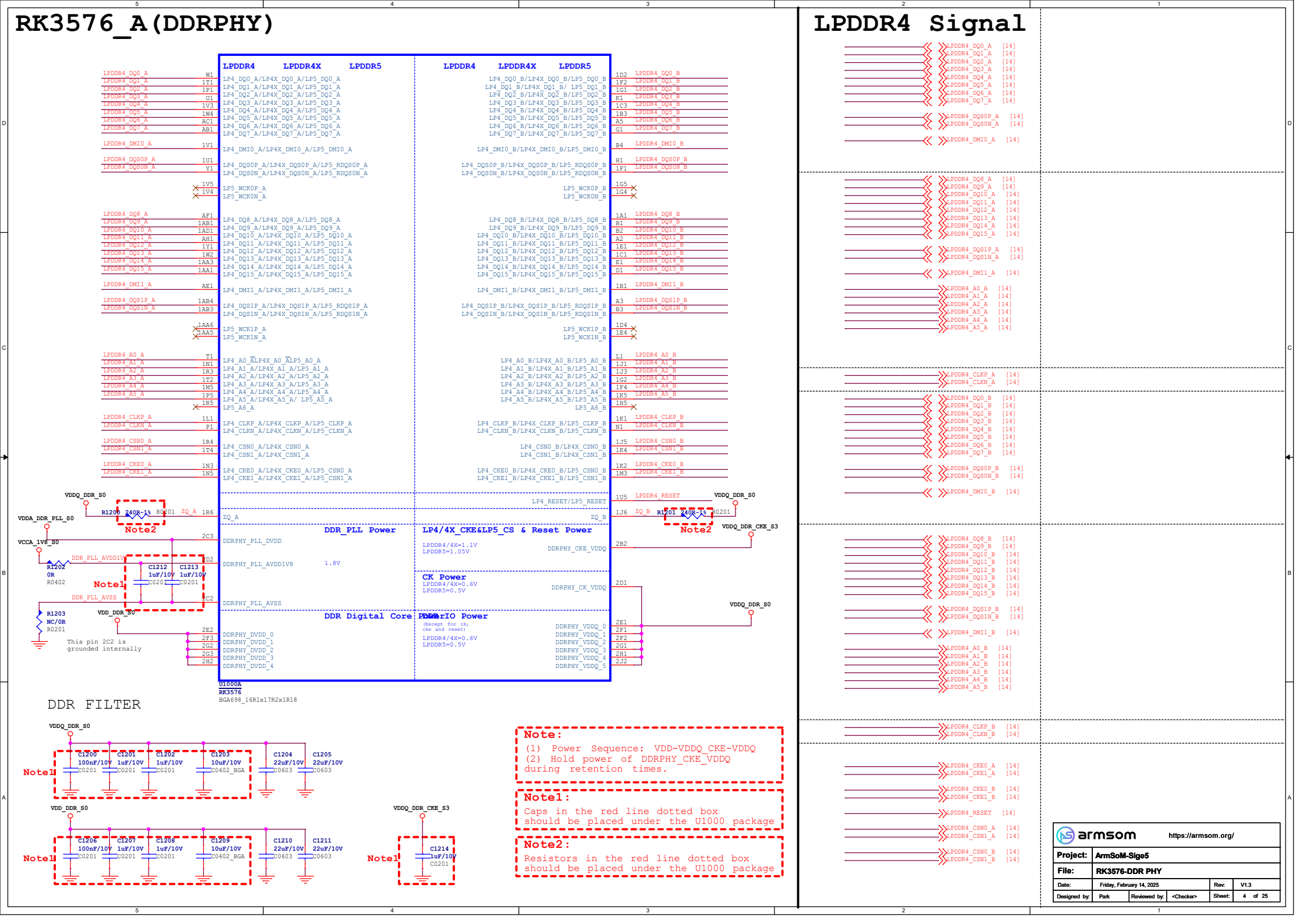
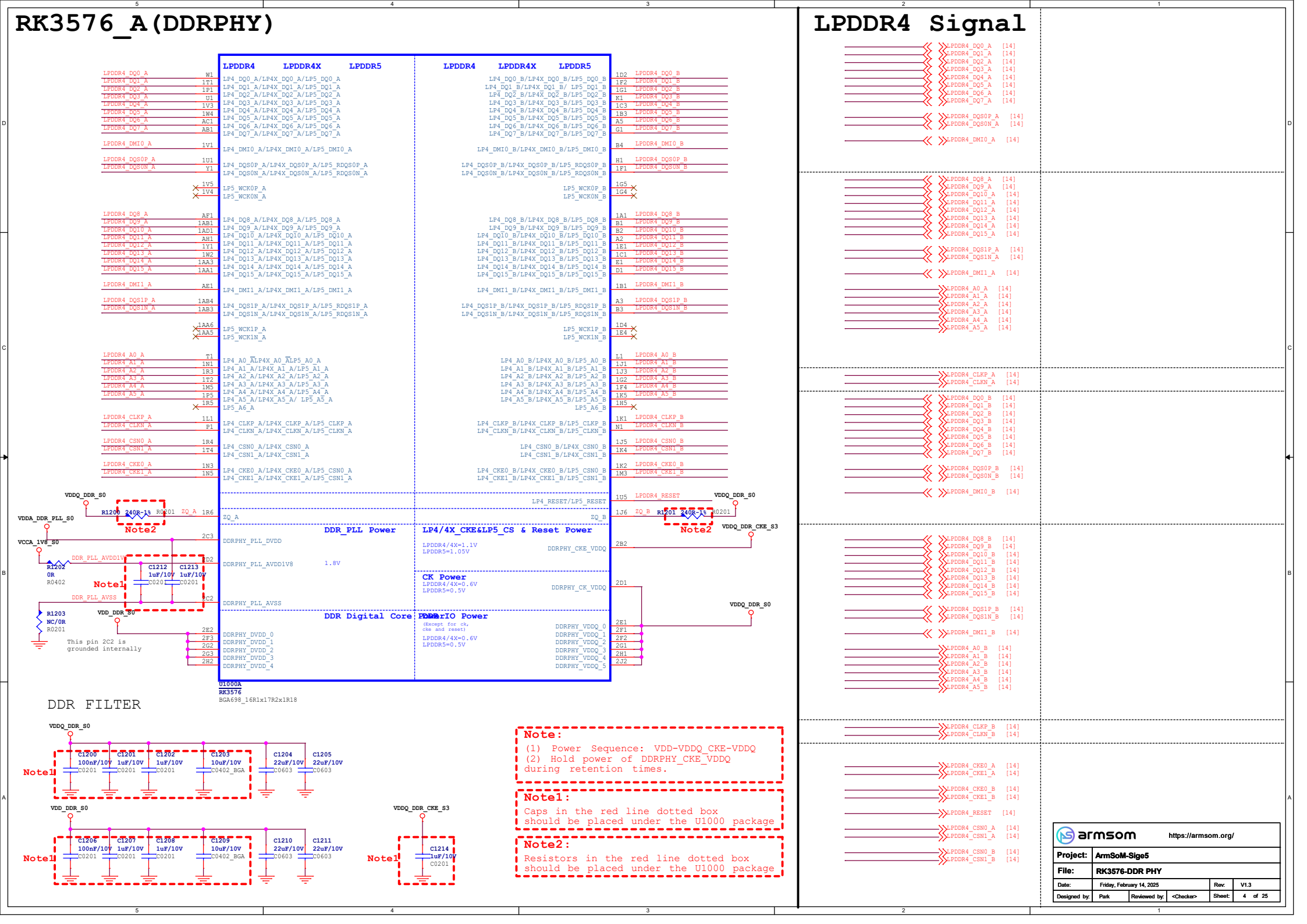
Date: Friday, February 14, 2025

Designed by: Park

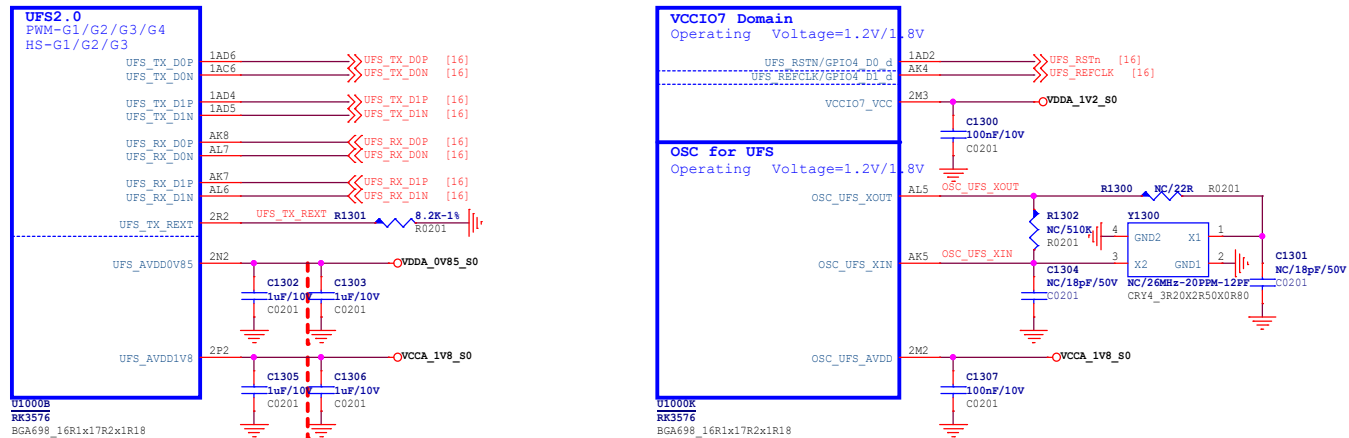
Reviewed by: <Checker>

Rev: V1.3

Sheet: 4 of 25

[illegible]

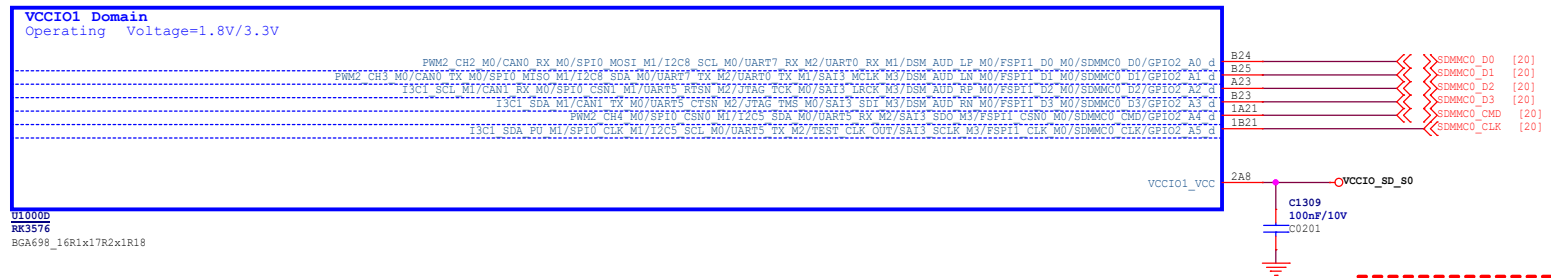
RK3576 B
(UFS2.1)



RK3576 C
(VCCI00)



RK3576 D
(VCCI01)

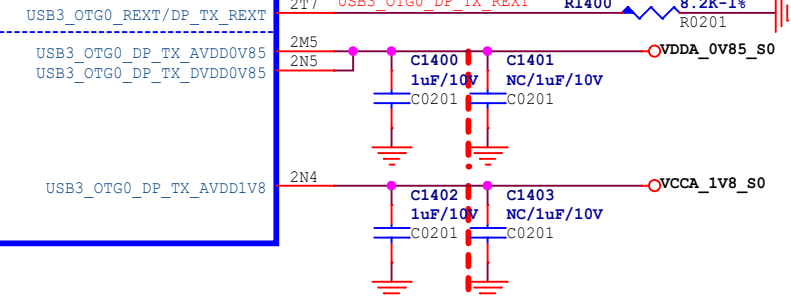
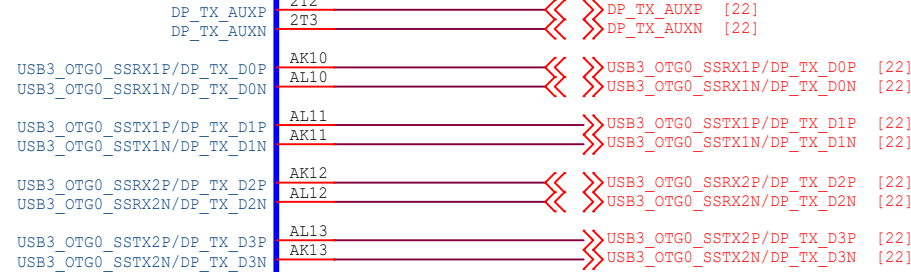


Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3576 L (USB3/DP)

USB3 OTG0/DP1.4 Alt
USB:USB3.2 Gen1x1 OTG0
DP :RBR/HBR/HBR2/HBR3

Support:
Type-C With Displayport Alternate Mode



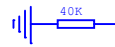
U1000L
RK3576
BGA698 16R1x17R2x1R18

Note:

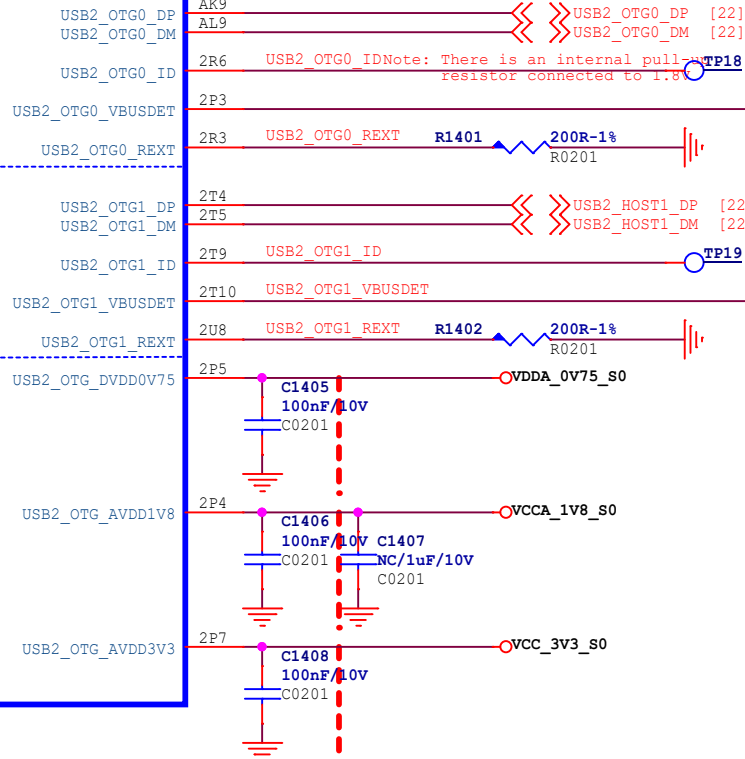
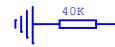
Caps of between dashed red lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

RK3576 M (USB2)

USB2 OTG0
OTG/HOST/DEVICE
HS/FS/LS Download Port



USB2 OTG1
OTG/HOST/DEVICE
HS/FS/LS



U1000M
RK3576
BGA698 16R1x17R2x1R18

The USB2 OTG1 function cannot be used, if the PCIe1 or SATA1 function of Combo PHY1 is selected

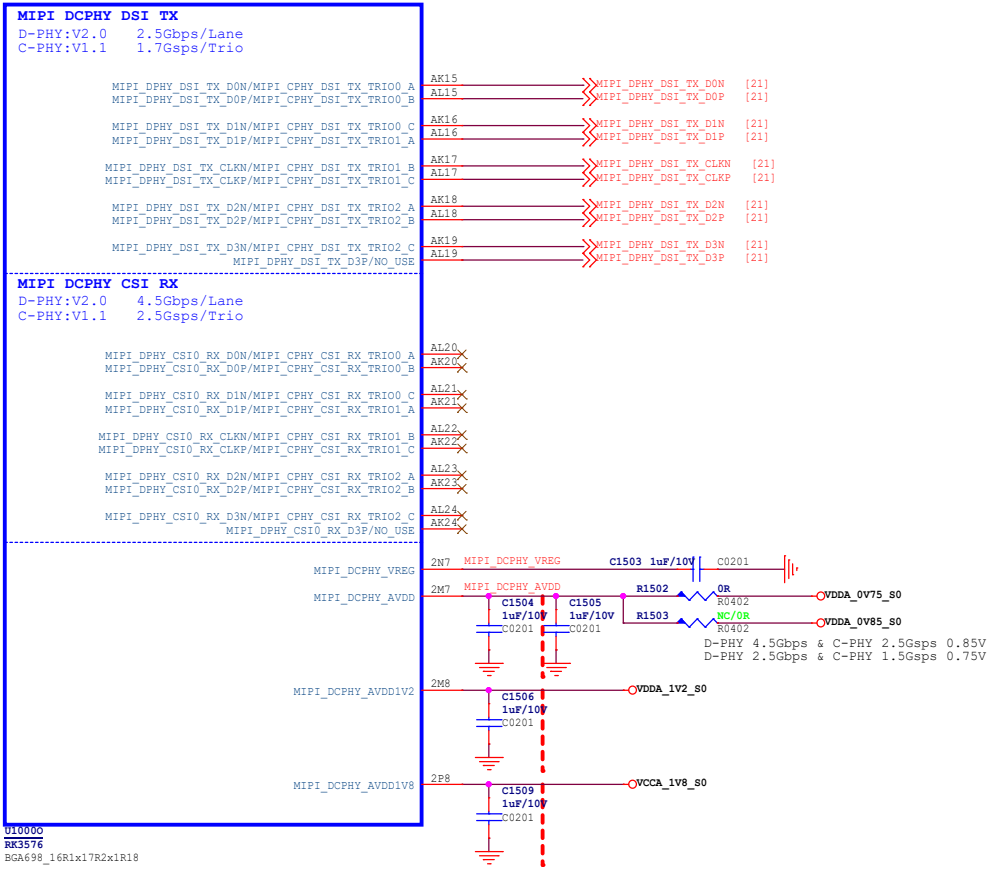


armsom

<https://armsom.org/>

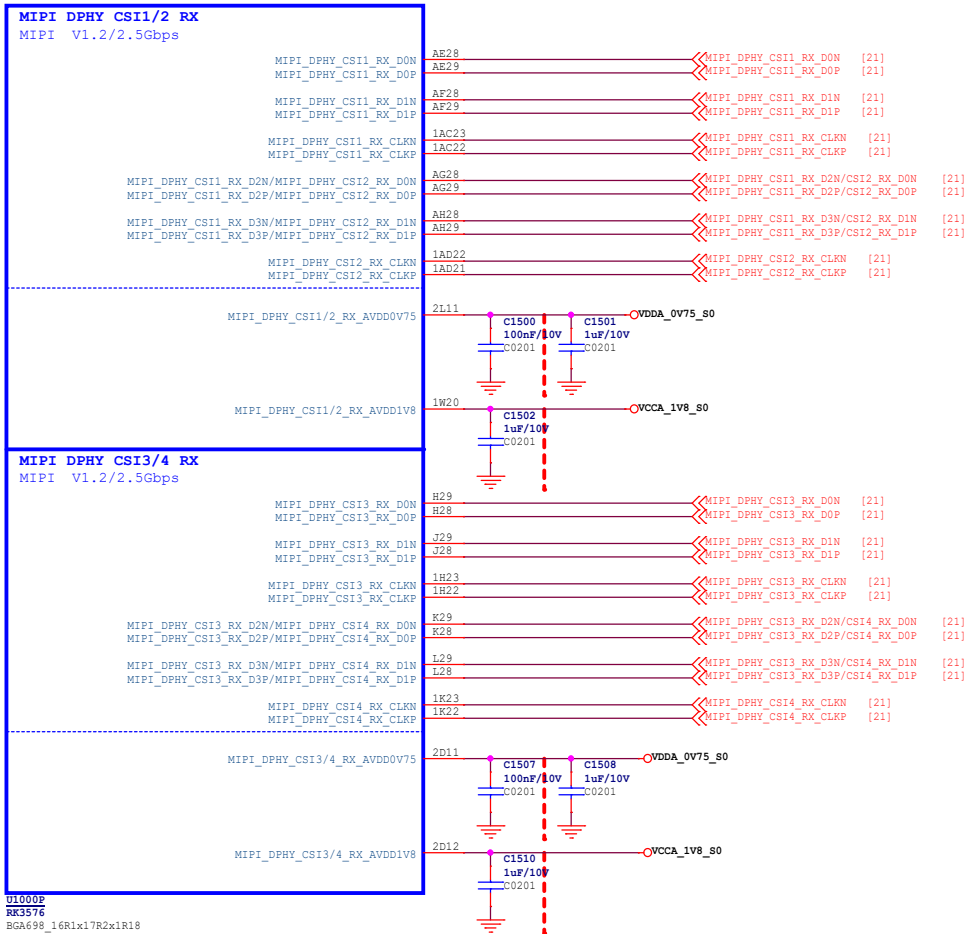
Project:	ArmSoM-Sige5				
File:	RK3576-TypeC/USB				
Date:	Friday, February 14, 2025			Rev:	V1.3
Designed by:	Park	Reviewed by:	<Checker>	Sheet:	6 of 25

RK3576_O (MIPI DCPHY)



Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

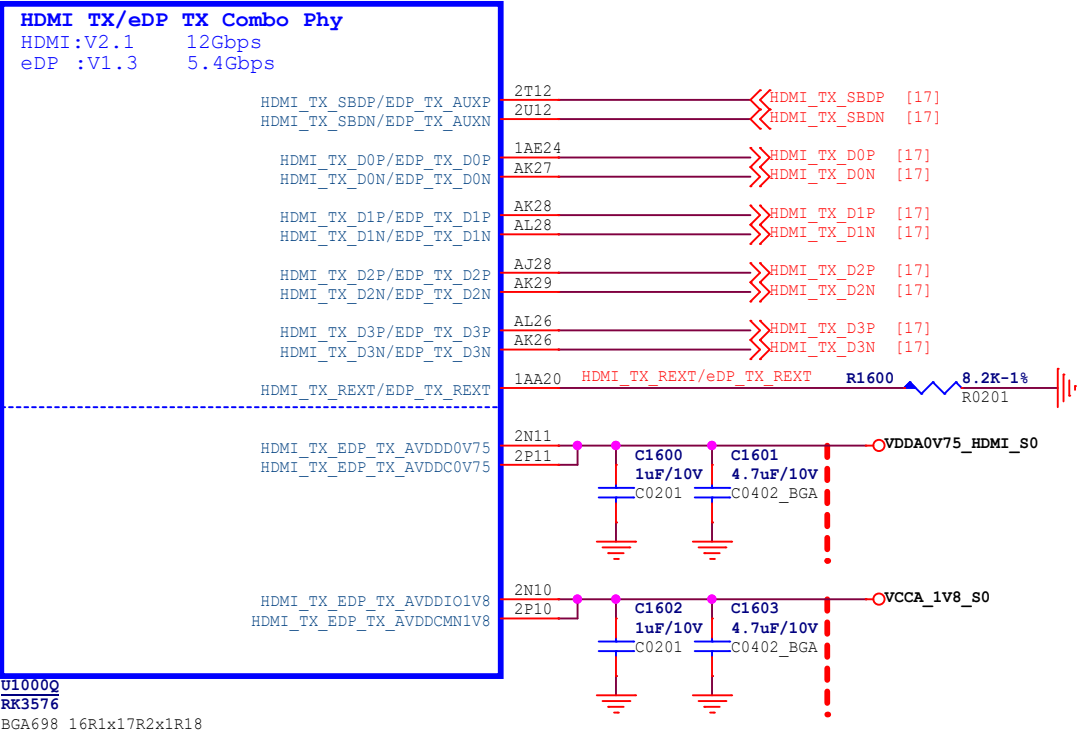
RK3576_P (MIPI DPHY CSI RX)



Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3576_Q (HDMI/eDP)

Note:
HDMI 2.1 supports up to 4Kx2K@120Hz



Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

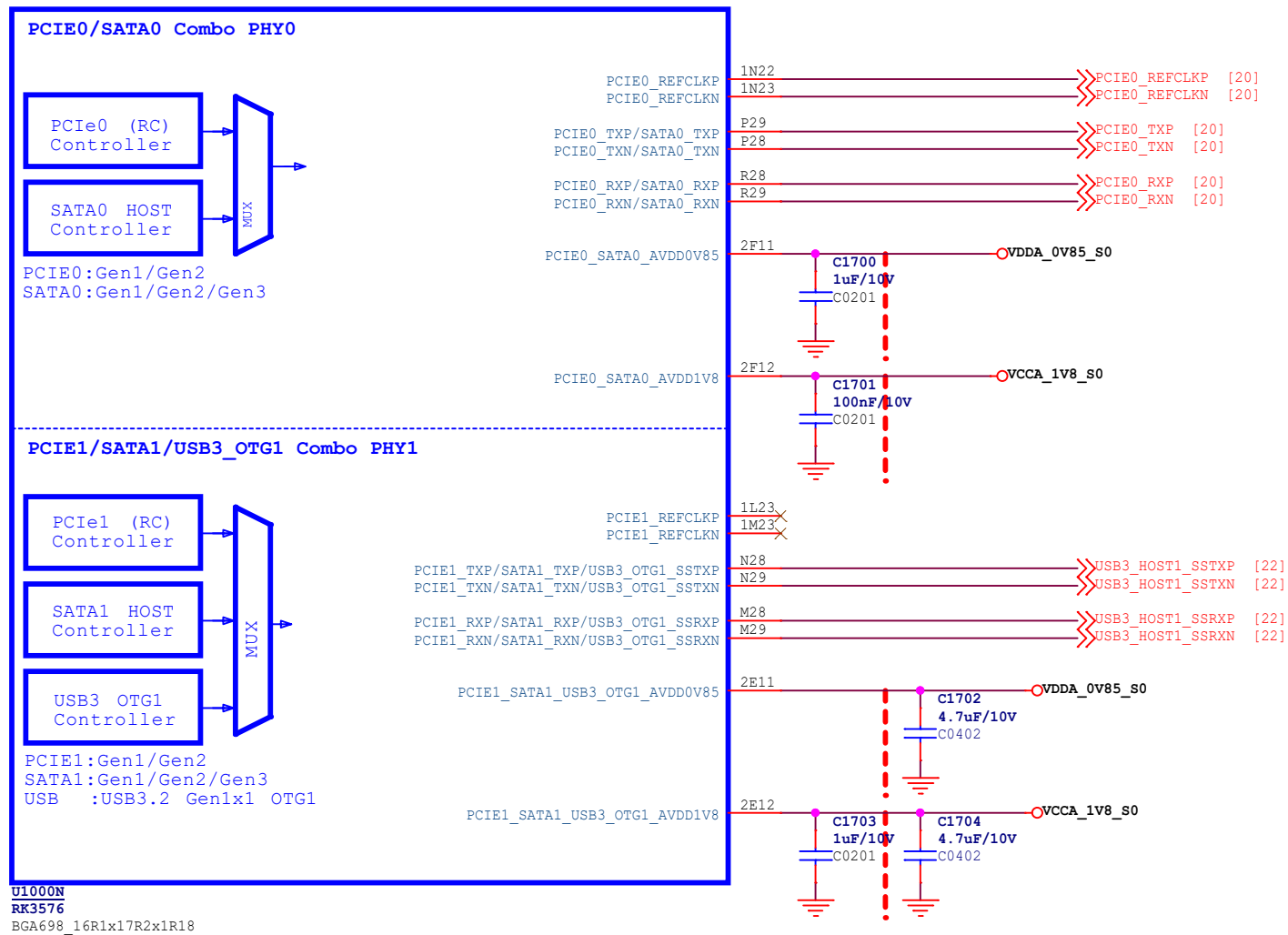


armsom

<https://armsom.org/>

Project:		ArmSoM-Sige5	
File:		RK3576-MIPI DSI/CSI	
Date:	Friday, February 14, 2025		Rev: V1.3
Designed by:	Park	Reviewed by:	<Checker>
Sheet:		8 of 25	

RK3576_N (PCIe/SATA/USB3)

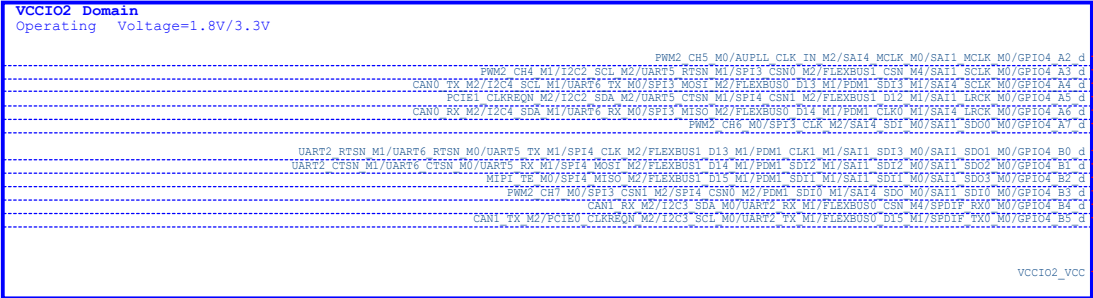


Note:

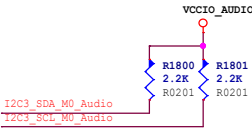
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

		https://armsom.org/	
Project:	ArmSoM-Sige5		
File:	RK3576-PCIe/SATA/USB3		
Date:	Friday, February 14, 2025		Rev: V1.3
Designed by:	Park	Reviewed by:	<Checker>
Sheet:	9		of 25

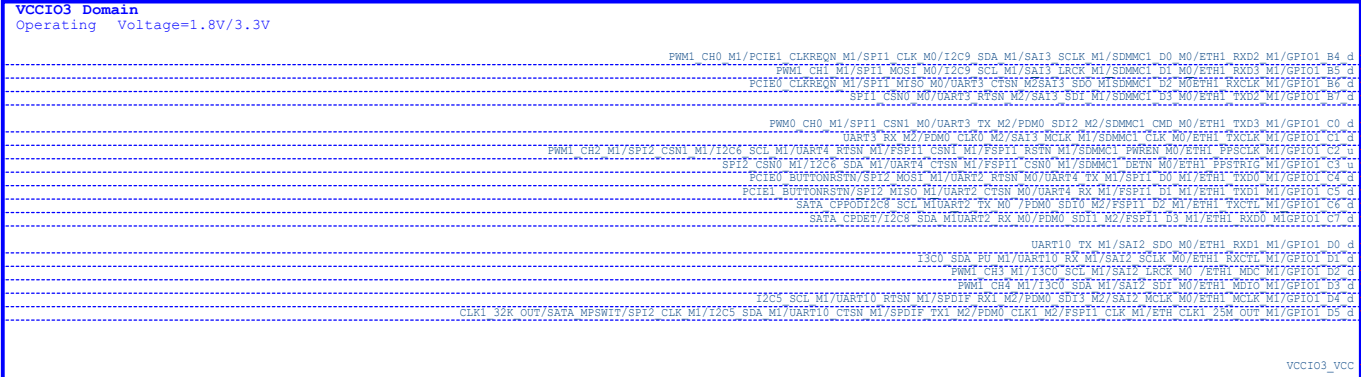
RK3576_F (VCCIO2)



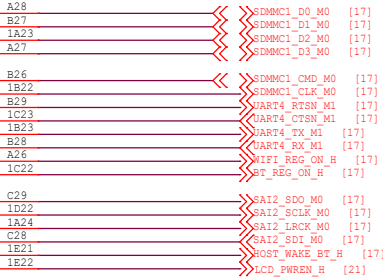
U1000F
RK3576
BGA698_16R1x17R2x1R18



RK3576_G (VCCIO3)

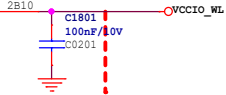


U1000F
RK3576
BGA698_16R1x17R2x1R18

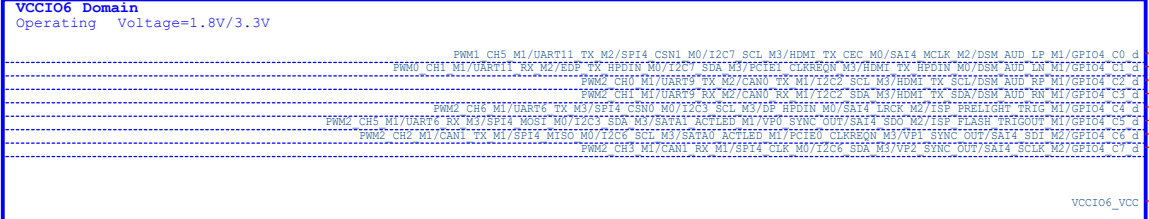


For BT

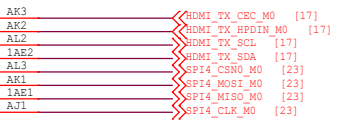
For BT



RK3576_J (VCCIO6)



U1000J
RK3576
BGA698_16R1x17R2x1R18

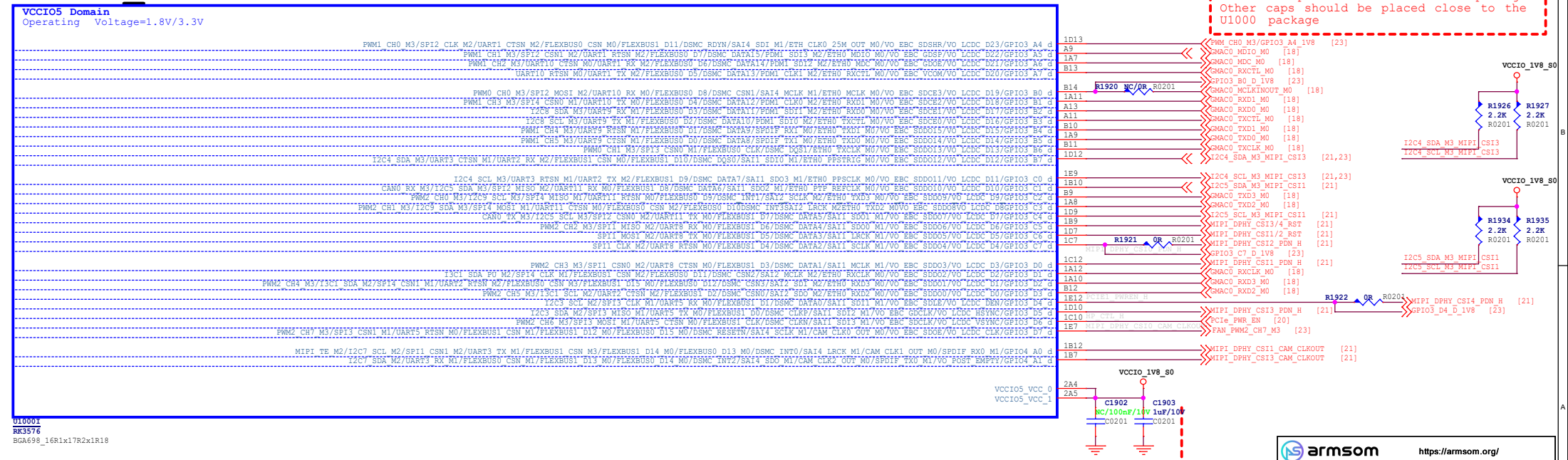


 https://armsom.org/	
Project:	ArmSoM-Sig5
File:	RK3576-GPIO VCCIO2/3/6
Date:	Friday, February 14, 2025
Designed by:	Park
Reviewed by:	<Checker>
Rev:	V1.3
Sheet:	10 of 25

RK3576_H (VCCIO4)



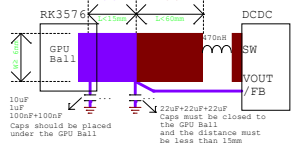
RK3576_I (VCCI05)



Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

PMIC RK806S-5 BUCK

- (3) I2C1_SDA_M0_RK806S
- (3) I2C1_SCL_M0_RK806S
- (3) PMIC_PWR_CTRL1
- (3) PMIC_PWR_CTRL2
- (3) PMIC_INT_1
- (3,23) RESET_1
- (13) PMIC_EXT_EN_OUT
- (23) PMICOK_1



Default:0.61V

DOB Type	Voltage	Rs
LPDS084/43	0.61V	22K 1%
LPDS085	0.51V	2K 1%

Default:1.1V

DOB Type	Voltage	Rs
LPDS084/43	1.1V	120K 1%
LPDS085	1.05V	210K 1%

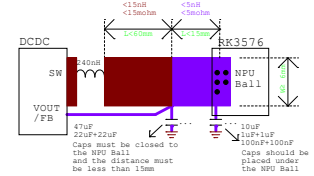
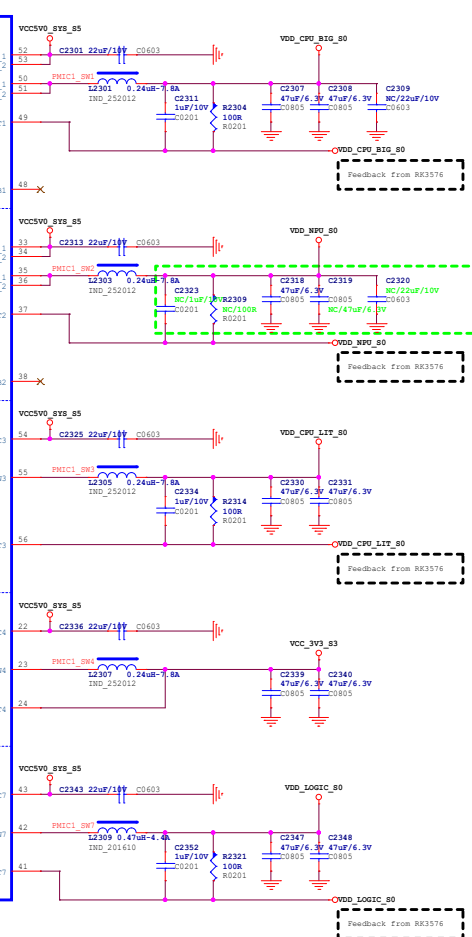
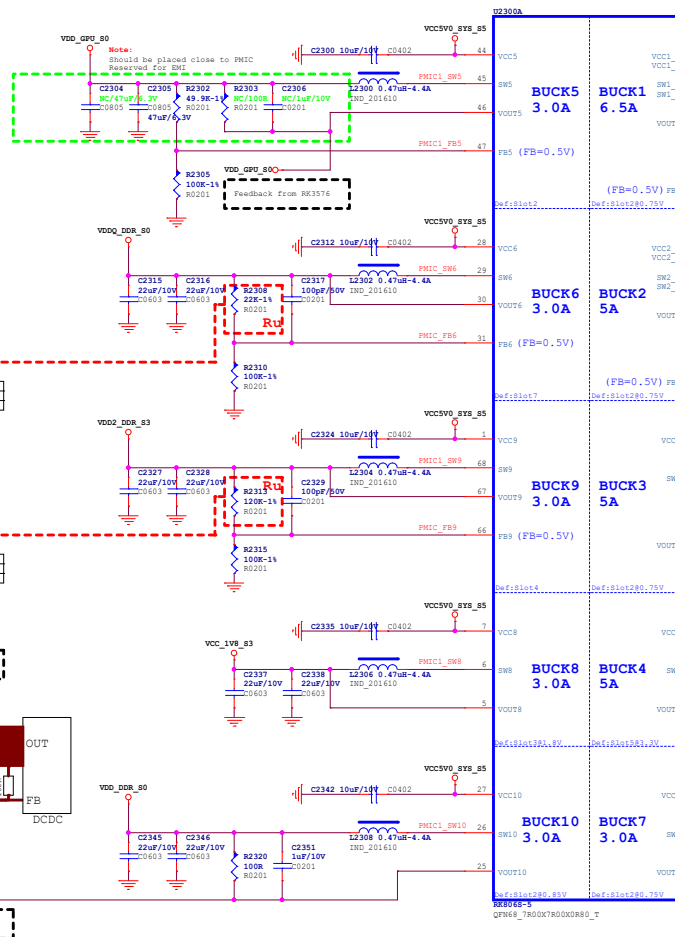
Default:1.8V



Default:0.85V

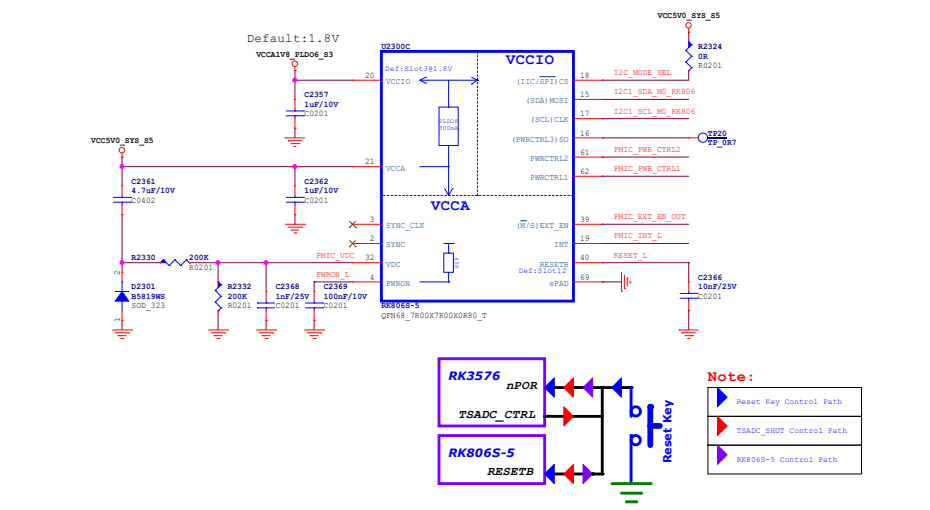
Low frequency
Operating Supply Voltage +5.5V(5.25-6V)
Peak Pulse Current>10A(90ms/20us)
Surge Clamping Voltage<6.5V

DO NOT DELETE IT!

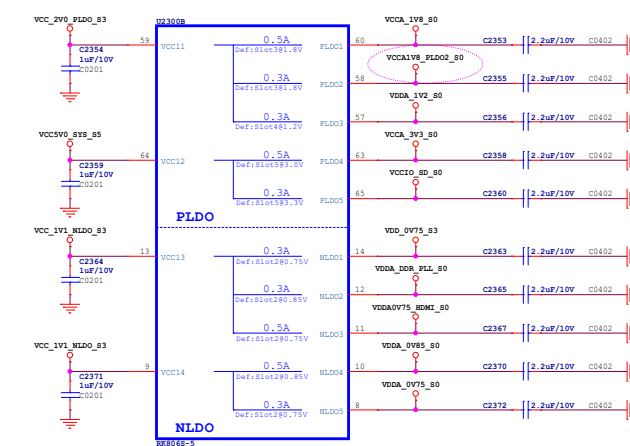


PMIC RK806S-5 Management

Note:
I2C Mode:CS(pin18) connected to VCCA(pin21);
SPI Mode(Def):CS(pin18) floating or connected to GND



PMIC RK806S-5 LDO



Default:1.8V

Default:1.8V

Default:1.2V

Default:3.0V

Default:3.3V

Default:0.75V

Default:0.75V

Default:0.85V

Default:0.75V

Default:0.75V

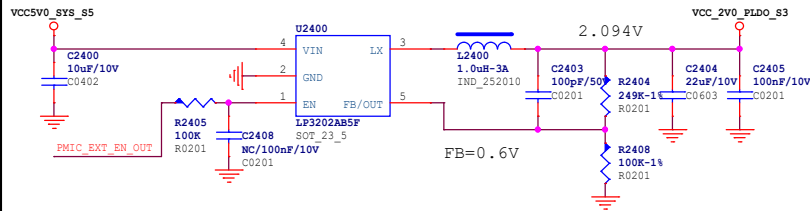
Default:0.75V

Default:0.75V

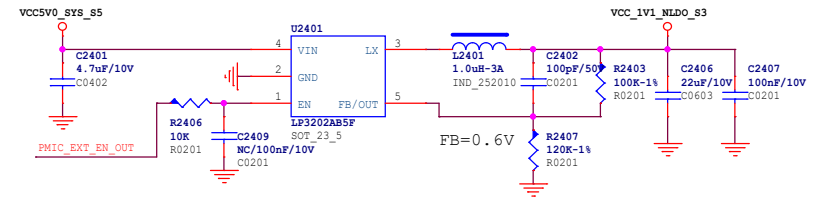
Default:0.75V

Note:
The RK806 LDO power distribution of the reference schematics is only suitable for the interface used in the reference schematics. If other interface functions are to be added to the reference schematics, the RK806 LDO distribution needs to be re evaluated, otherwise the added functions may exceed the maximum current provided by the LDO

VCC_2V0_PLDO_S3

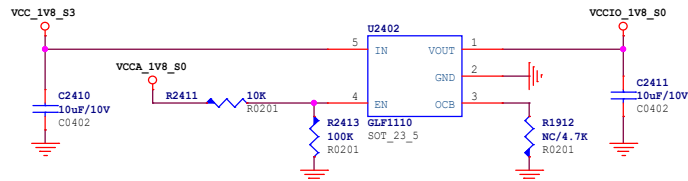


VCC_1V1_NLDO_S3



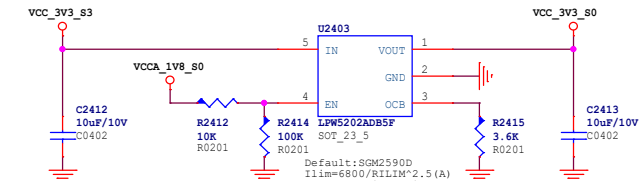
VCCIO_1V8_S0

Note: Need quick output discharge

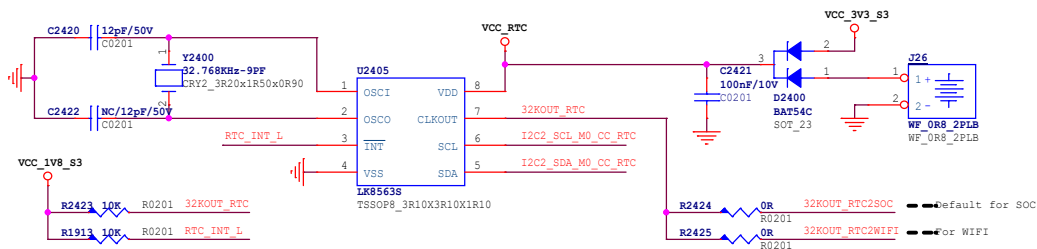


VCC_3V3_S0

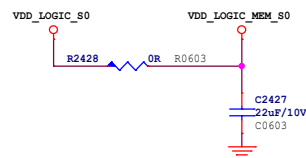
Note: Need quick output discharge



RTC



VDD_LOGIC_MEM EXT (Option for test)



		https://armsom.org/	
Project:	ArmSoM-Sig5		
File:	Power-Ext Discrete/RTC		
Date:	Friday, February 14, 2025		Rev: V1.3
Designed by:	Park	Reviewed by:	<Checker>
Sheet:	13 of 25		

eMMC FLASH

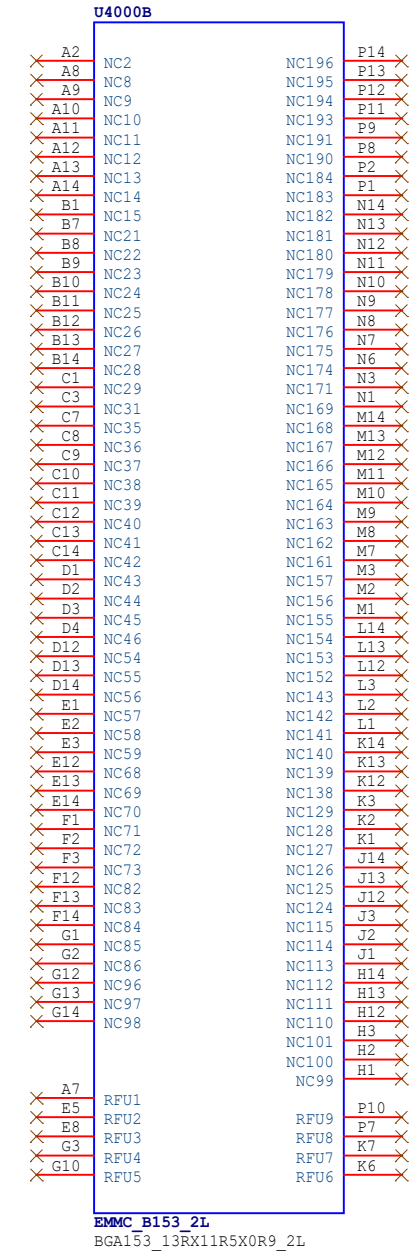
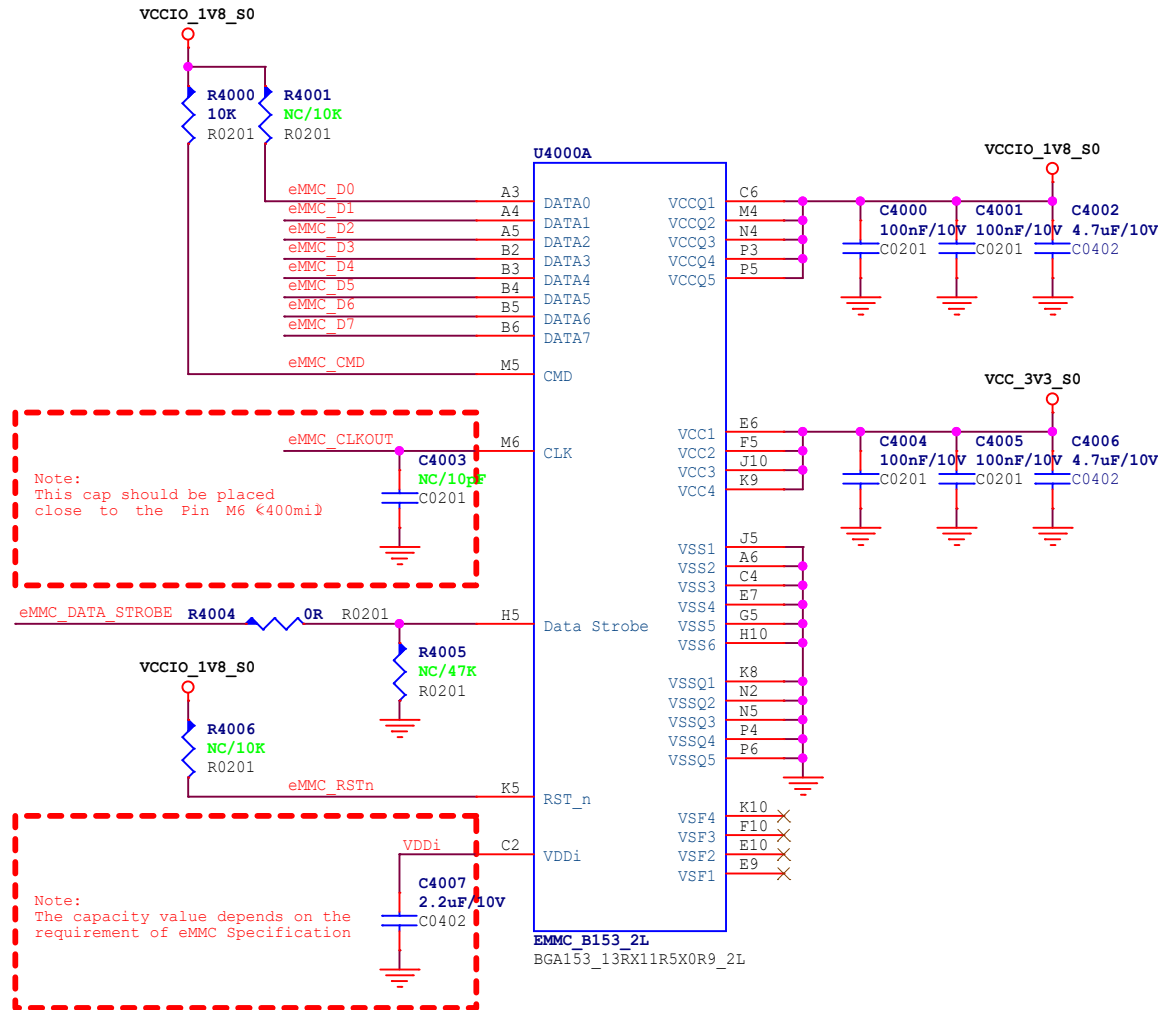
[5] eMMC_D0<<>>
[5] eMMC_D1<<>>
[5] eMMC_D2<<>>
[5] eMMC_D3<<>>
[5] eMMC_D4<<>>
[5] eMMC_D5<<>>
[5] eMMC_D6<<>>
[5] eMMC_D7<<>>

[5] eMMC_CMD<<>>

[5] eMMC_CLKOUT<<>>

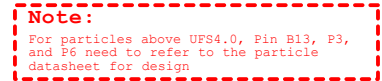
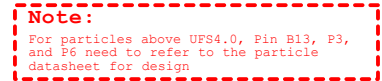
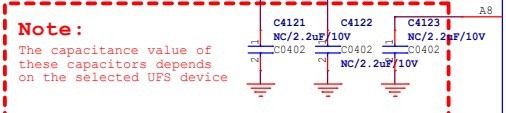
[5] eMMC_DATA_STROBE<<>>

[5] eMMC_RSTn<<>>



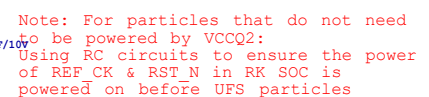
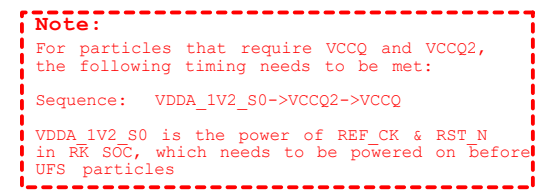
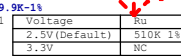
		armsom		https://armsom.org/	
Project:		ArmSoM-Sige5			
File:		Flash-eMMC			
Date:		Friday, February 14, 2025		Rev:	V1.3
Designed by:		Park	Reviewed by:	<Checker>	Sheet: 15 of 25

[5]	UFS_TX_D0P	
[5]	UFS_TX_D0N	
[5]	UFS_TX_D1P	
[5]	UFS_TX_D1N	
[5]	UFS_RX_D0P	
[5]	UFS_RX_D0N	
[5]	UFS_RX_D1P	
[5]	UFS_RX_D1N	
[5]	UFS_RSTn	
[5]	UFS_REFCLK	

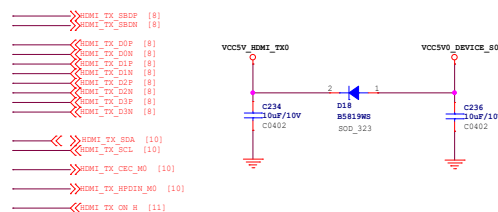


	VCCQ	VCCQ2	VCC	Default UFS device: UFS2.2
UFS2.0	1.2V	1.8V	3.3V	
UFS2.1	No Connect	1.8V	3.3V	
UFS2.2	No Connect	1.8V	3.3V	
UFS3.0	1.2V	No Connect	2.5V/3.3V	
UFS3.1	1.2V	No Connect	2.5V/3.3V	
UFS4.0	1.2V	No Connect	2.5V	

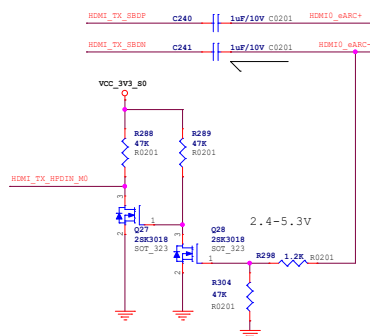
Note: Do not support UFS4.0 Device!
The power ball that is not used at the particle must be kept floating.



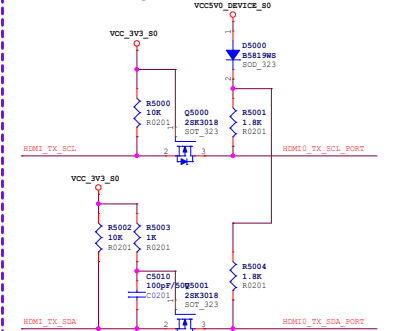
HDMI TX0



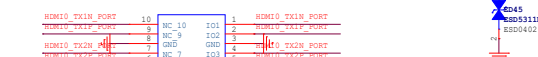
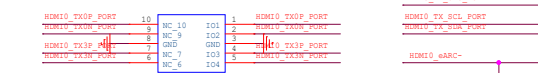
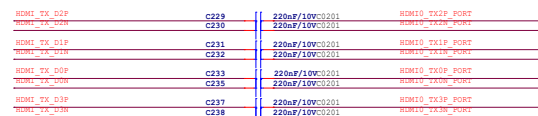
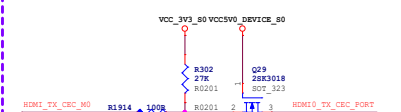
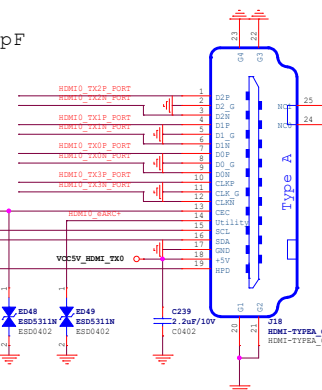
HDMI TX eARC



HDMI TX DDC



HDMI TX CEC

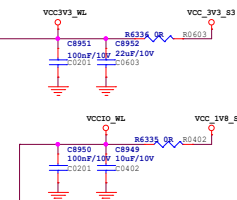
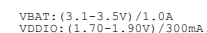
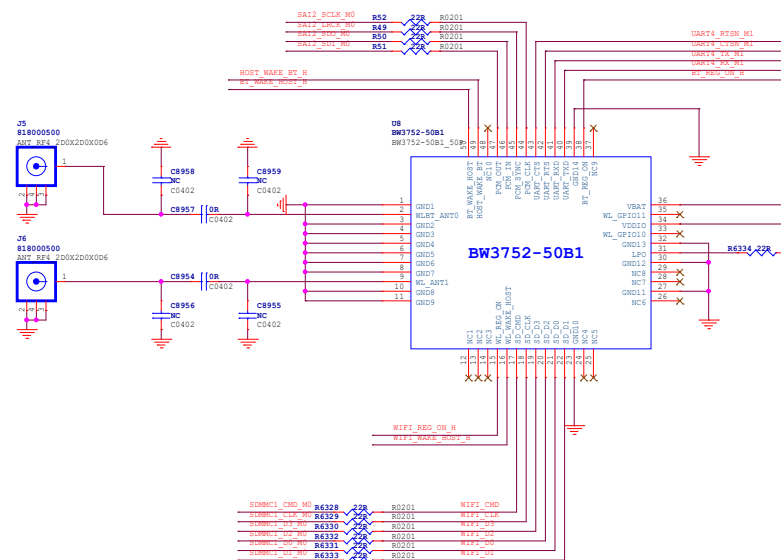
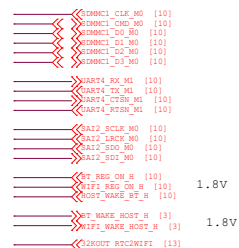
 $C_j \leq 0.2 \text{ pF}$ 

Note:
The HDMI.2.1 trace length is less than 100mm.
The HDMI.2.1 differential trace impedance is 100 OHM

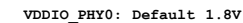
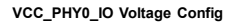
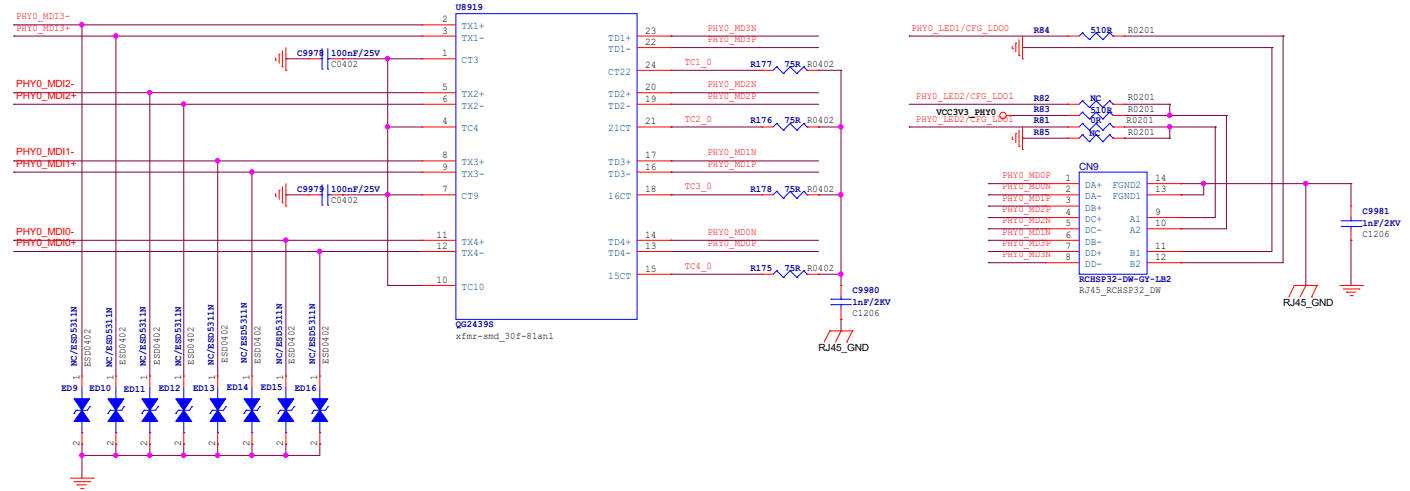
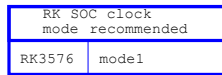
Note:
The controller only support AC coupled link. In order to backward compatibility, or to meet HDMI2.0 (1.4b) DC common mode spec and Voff, need do R based level-shift.

```
Switch on in HDMI2.0(TMDS) mode
Switch off in HDMI2.1(FRL) mode.
```

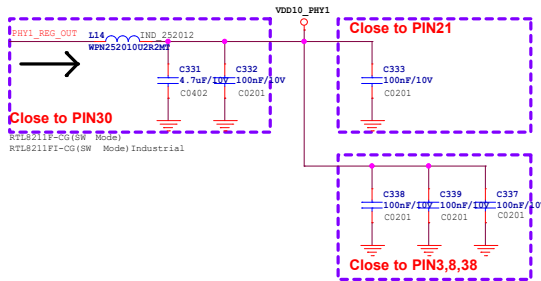
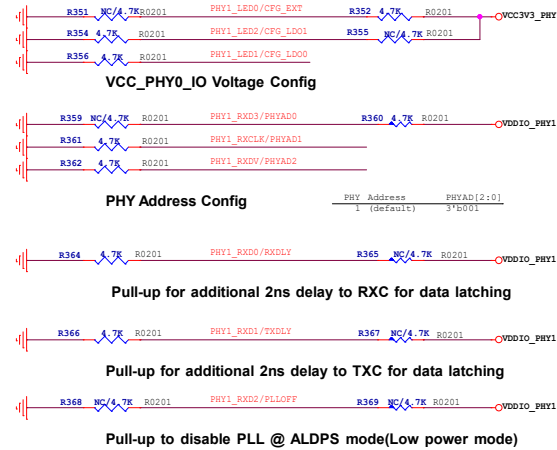
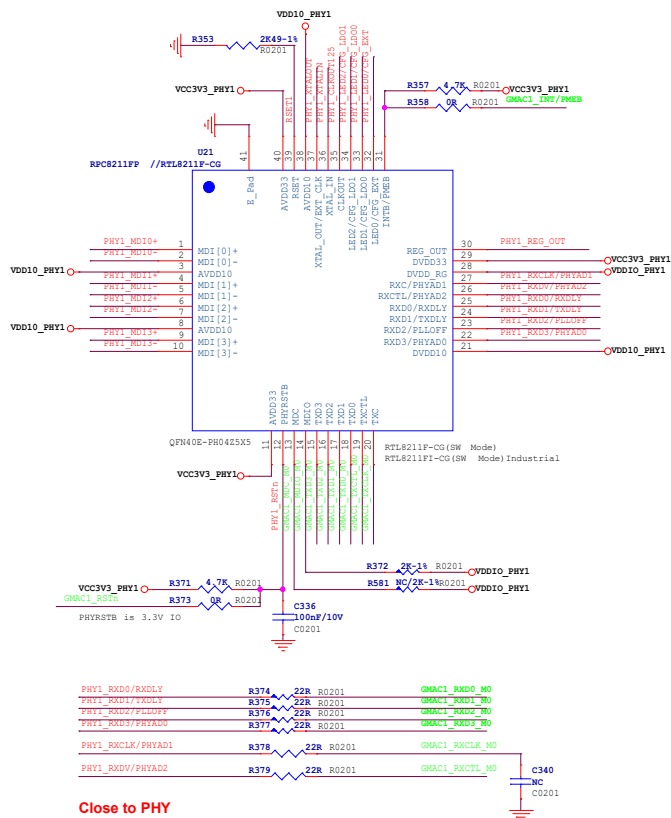
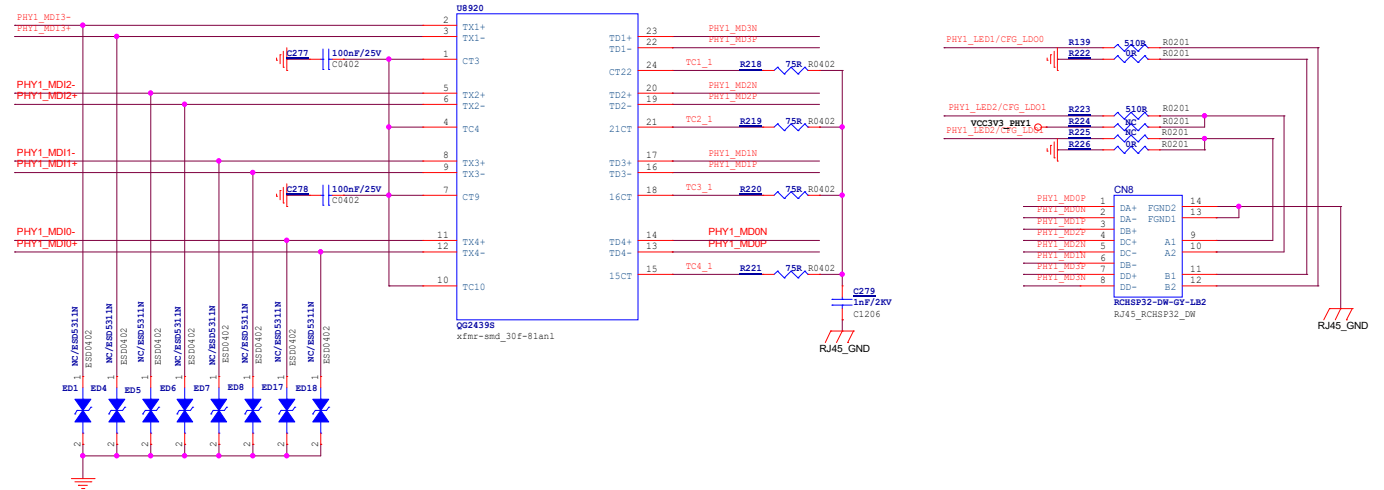
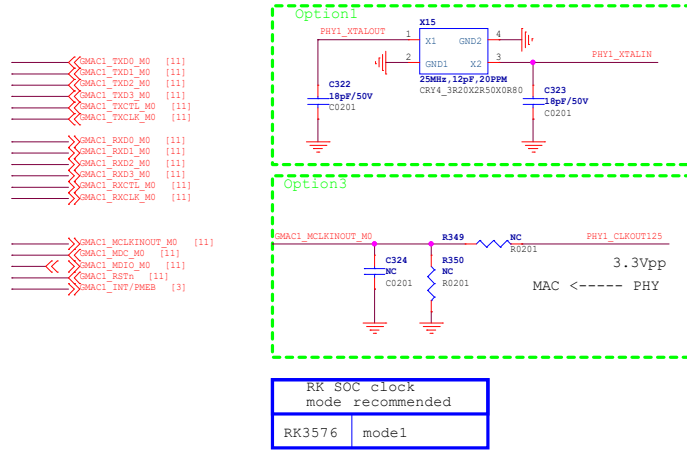
SDIO WIFI6/BT Module-2T2R



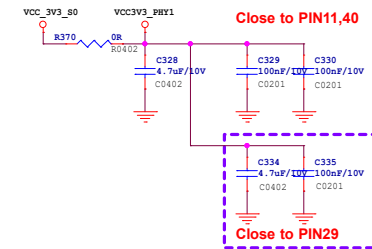
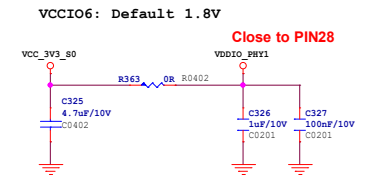
	>>GNACO_TXD0_M0	[11]
	>>GNACO_TXD1_M0	[11]
	>>GNACO_TXD2_M0	[11]
	>>GNACO_TXD3_M0	[11]
	>>GNACO_TXCTL_M0	[11]
	>>GNACO_TXCLK_M0	[11]
	>>GNACO_RXD0_M0	[11]
	>>GNACO_RXD1_M0	[11]
	>>GNACO_RXD2_M0	[11]
	>>GNACO_RXD3_M0	[11]
	>>GNACO_RXCTL_M0	[11]
	>>GNACO_RXCLK_M0	[11]
	>>GNACO_MCLKINOUT_M0	
	>>GNACO_MDC_M0	[11]
<<	>>GNACO_MDIO_M0	[11]
	>>GNACO_RSTn	[11]
	>>GNACO_INT/PMBE	[11]

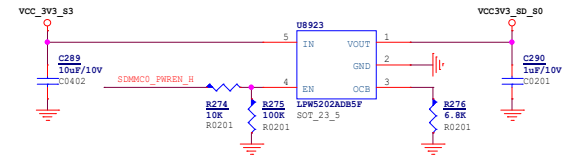
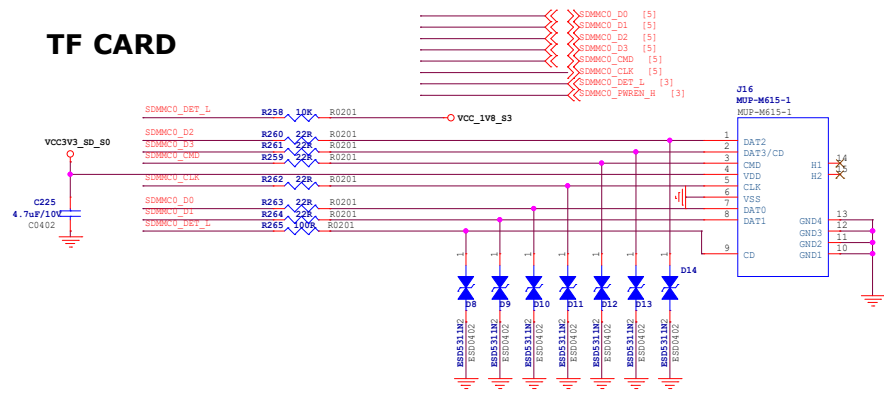


Giga PHY1_WAN

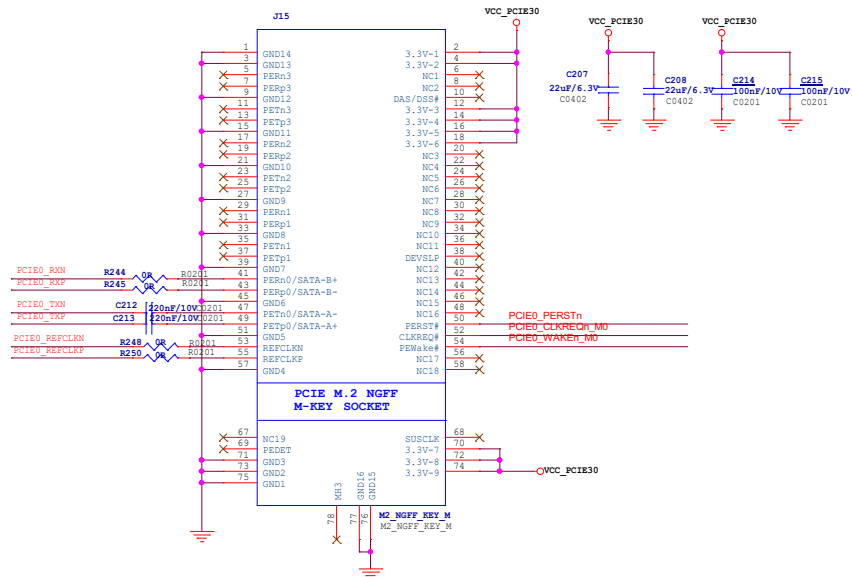
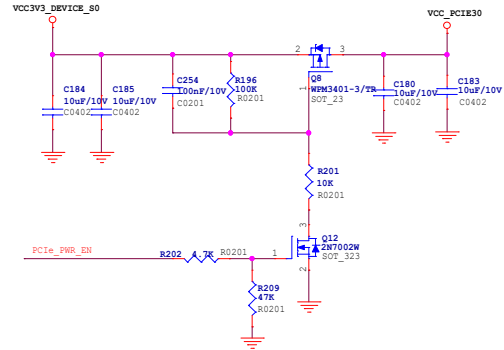
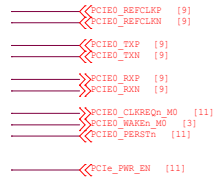


RGMI Power Source	CFG EXT	CFG LDO[1:0]
External 3.3V(default)	1'b1	2'b00
External 1.8V	1'b1	2'b10
Internal 1.8V	1'b0	2'b10

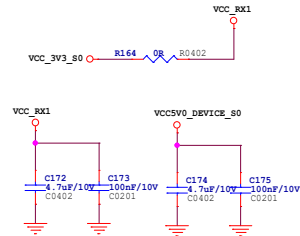
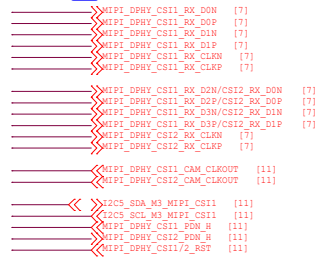


TF CARD

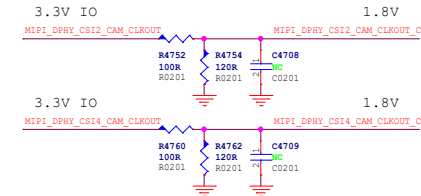
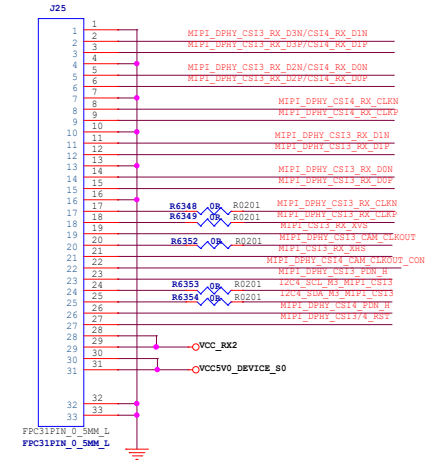
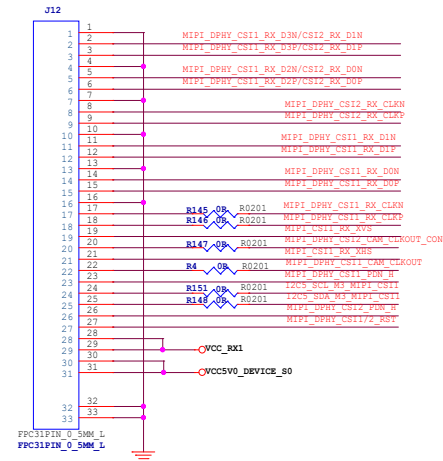
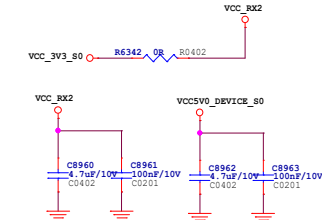
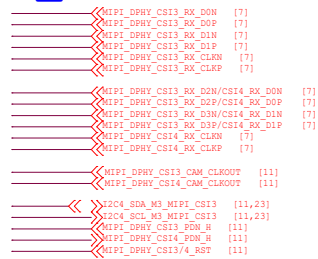
M.2_PCIe



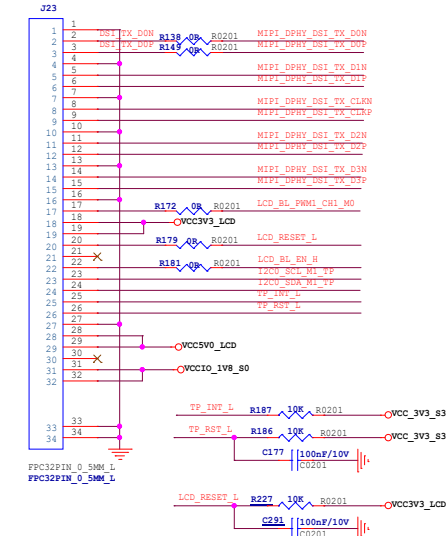
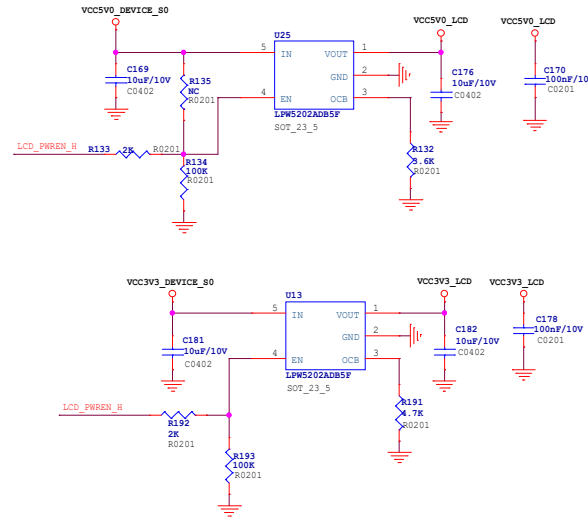
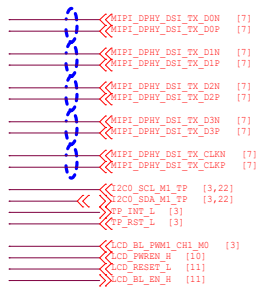
CSI0_MIPI



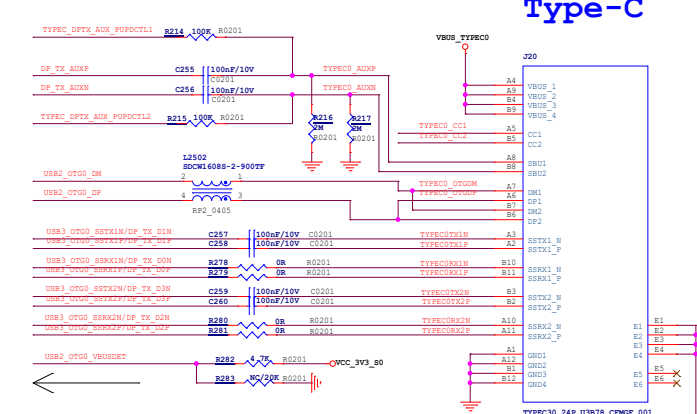
CS1_MIPI



DSI_MIPI



DCIN



VCC_SYSIN

5V
3.3V

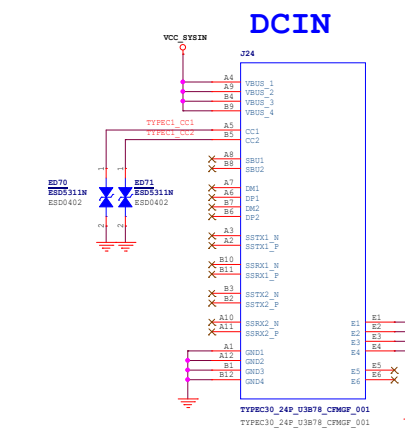
R6388
100K, 1%
R6201

R6390
2K, 1%
R6201

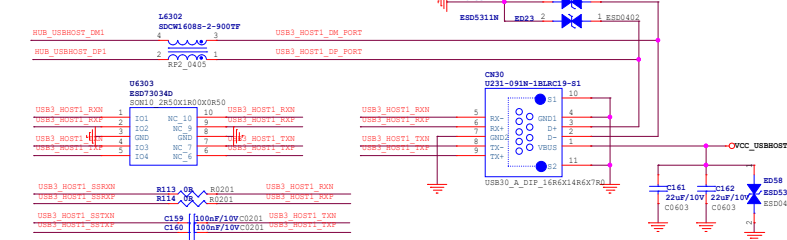
R6386
10K, 1%
R6201

PARADC_VIN7

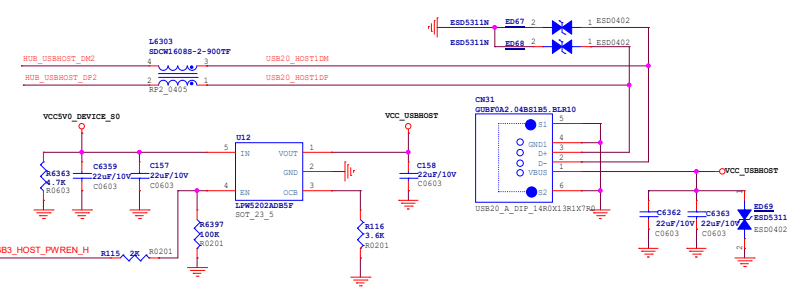
TYPEC_IN	12Bit_ADC	ADC6
5V	35E	0.37V
9V	610	0.68V
12V	815	0.9V
15V	A1A	1.136V
20V	D79	1.51V



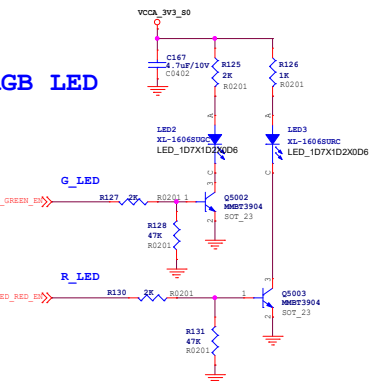
USB3.0



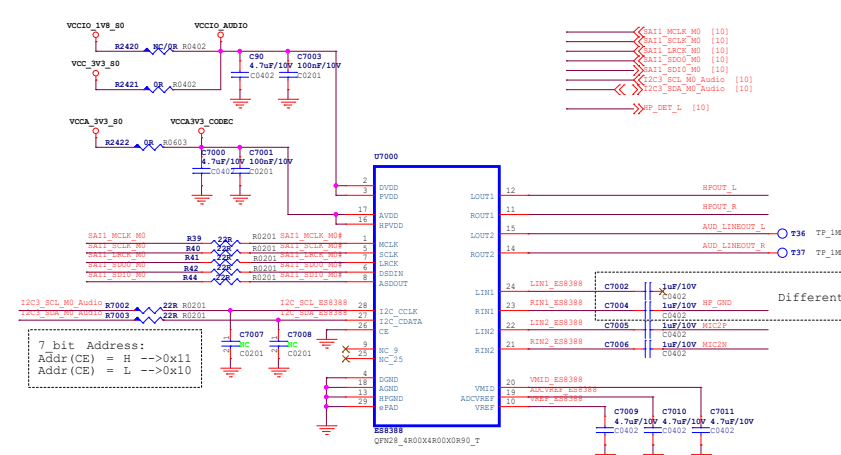
USB2.0



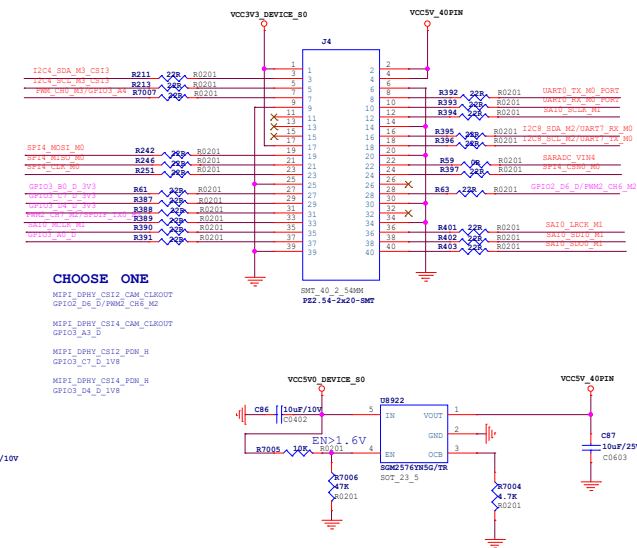
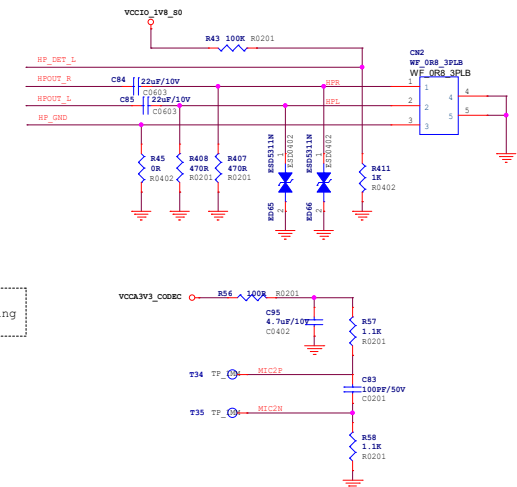
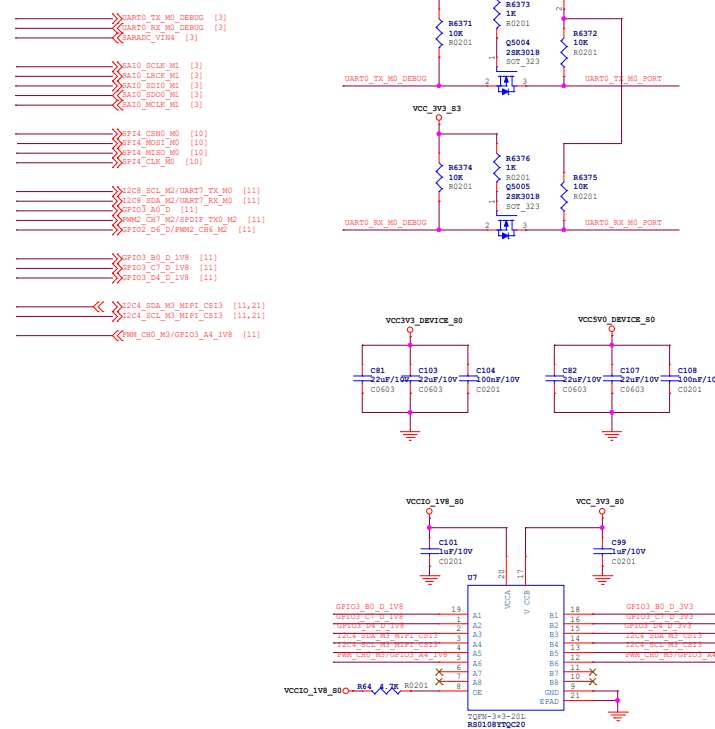
RGB LED

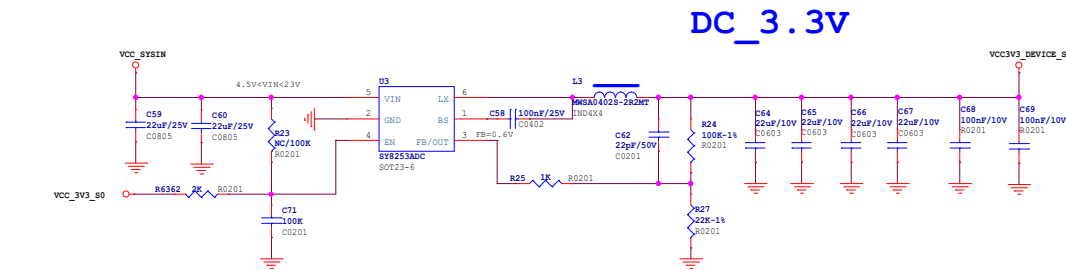
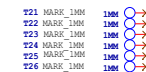
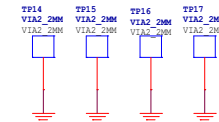
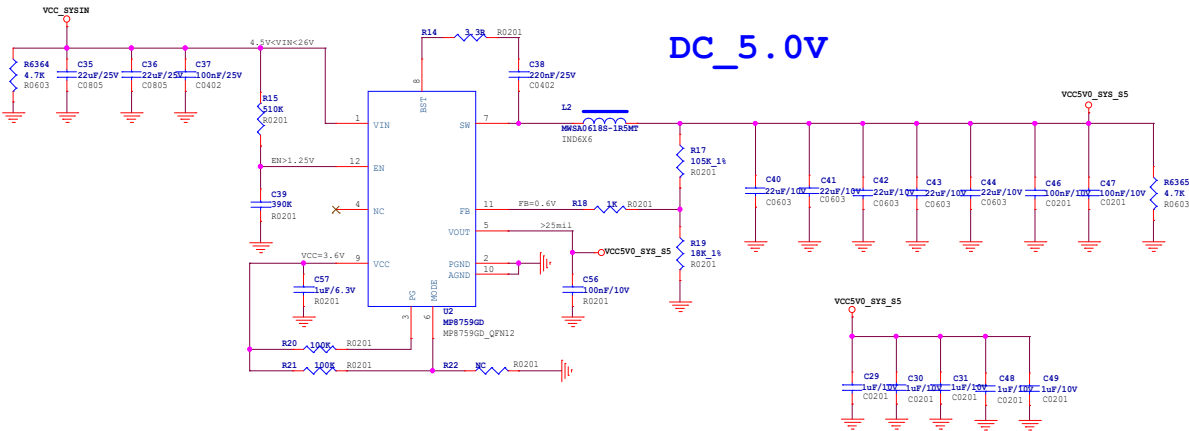
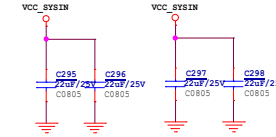
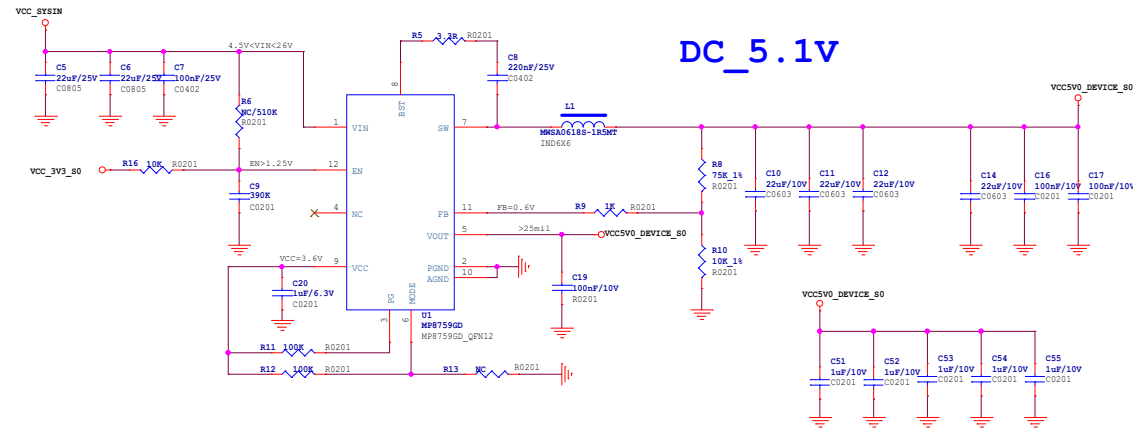


AUDIO CODEC



40PIN_GPIO





Revision History

Version	Date	By	Change Dscription	Approved
V1.0	2024-03-27	SL Chen	First release;	
V1.1	2024-05-15	SL Chen	1.U1/U2 Pin5 connect to output; TF_DET_L connect to VCC_1V8_S3; J23 Pin2&Pin3 change position; 2.J25 MIPI_DPHY_CSI3_CAM_CLKOUT&MIPI_DPHY_CSI4_CAM_CLKOUT_CON change position;	
V1.2	2024-07-01	SL Chen	Wifi Module=BL-M8852BS2 change to BW3752-50B1;	
V1.3	2025-01-21	SL Chen	ADC Power=VCCA1V8_S0 change to VCCA1V8_PLDO2_S0;	